

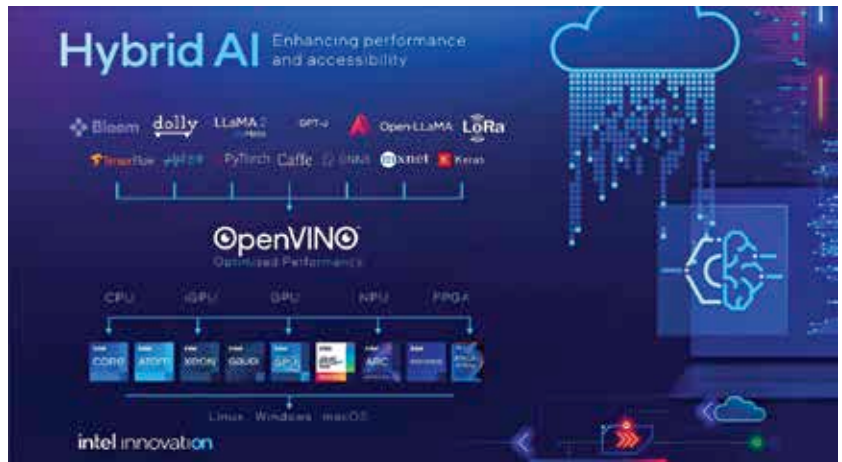
allow for Hybrid AI use cases to grow, where larger models and training still happens in the cloud but AI inference and smaller models spring into action at a local level. Secondly, where the cloud AI models are vast and segregated (running into a trillion parameters or more), on the edge devices will need far more integrated AI models (below 30 billion parameters), where AI has to be integrated into the hardware and software, according to Sachin Katti.

3 Open source software all the way

To quote Intel CEO Pat Gelsinger, “software defined, hardware enabled” is undoubtedly the way ahead. From the chipmaker’s perspective, as devices become more capable for running AI both locally and on the cloud, their efficiency will be ultimately unlocked by software that runs on top. Developers will play a crucial role in realizing this vision, for which Intel is busy creating the right incentives.

One of the key incentives for developers is OpenVINO – Open Visual Inference and Neural Network Optimization – Intel’s developer toolkit for faster inference of deep learning models. The latest 2023.1 version of OpenVINO demonstrated key optimizations to many GenAI models, including LLaMA by Meta. General availability of Intel Developer Cloud provides developers with access to Intel’s latest hardware, allowing them to not only build AI-optimized apps but also train and optimize AI models.

Another key piece is boosting the adoption of oneAPI, for developers deploying their codes on heterogeneous systems that use more than one kind of processor. Intel also spoke about Project Strata coming in 2024. According to Pat Gelsinger, Project Strata will play a key role in scaling infrastructure on the edge to enable Hybrid AI, giving developers key modular building blocks to better unlock an ecosystem of AI applications more efficiently going forward. At Intel Innovation, everyone from Red Hat to Arm approved of Intel’s focus on ensuring developers have access to tools and capabilities that allow them to code once and deploy anywhere, especially on the client and edge platforms.



4 AI competition with NVIDIA set to intensify

In the arena of AI, NVIDIA is the unquestioned heavyweight, a fact that even Intel acknowledges. But evidence is starting to emerge that Intel is finally gathering some momentum to challenge NVIDIA.

According to MLPerf benchmarks, an industry standard, Intel’s Gaudi2 came out ahead of NVIDIA’s A100/H100 in AI accelerated workloads for GPT-J, in terms of price to performance. In fact, Intel’s 4th gen Xeon platform delivered 99.9% accuracy for CPU only performance in GPT-J accelerated workload. Sure this is just a single workload on one specific benchmark, but it’s still a big validation, if you factor in the scale of Intel CPUs that power datacenters around the world – which is way larger in number than NVIDIA GPUs right now.

Alongside Intel Core Ultra for PCs and laptops, Intel’s 5th gen Xeon CPUs will also launch on December 14, 2023, while Gaudi3 is expected by early 2024. Whether it’s AI training or inference, high-performance computing or general cloud, Pat Gelsinger claims the company will deliver more TOPs (tera operations per second) than anyone else in the industry in 2024. Whether or not TOPs is just hype, one thing is clear: Intel’s claim to surpass even NVIDIA due to their sheer scale in the market should be a sign that the chipmaker has thrown down its gauntlet in the AI arena. Needless to say, competition between two big tech giants in the AI space is great for the ecosystem as a whole, whether it’s developers creating AI apps of tomorrow to customers and end

users like you and me that will unknowingly use them.

5 Chip-as-a-Service business model

While this isn’t about AI explicitly, Intel’s process node upgrades have far-reaching implications not just on the future of AI everywhere but also on the company’s own foundry business. Gelsinger’s extremely ambitious “5 nodes in 4 years” target remains on track, as the chipmaker gets back to chipmaking excellence with production of Intel 18A expected to begin in late 2024. Needless to say, Intel’s aggressive execution down the nanometer scale to the angstrom era will give the likes of TSMC and Samsung some serious pause.

If Intel matches up to TSMC and Samsung’s foundry capabilities in the next 12-18 months, it will allow them to entertain and manufacture chips for the likes of Apple, Qualcomm and even NVIDIA – all huge AI players and who don’t currently have their own chip manufacturing facilities. With a year still to go, quite a few industry analysts remain skeptical of Intel’s ambitious turnaround, of reaching the proverbial chipmaking promised land.

However, so far, under Pat Gelsinger’s talismanic leadership, Intel has met all the ambitious deadlines of the past 2 years that it set out for itself. With billions of dollars worth of new Intel fabs coming up in the US and Europe, in keeping pace with their strategic foundry ambitions, it’ll take a bold person to bet against Intel’s future which looks nothing short of exciting right now. **[E]**